Online appendix

Sensitivity analysis

One of the risks of using a machine-learning classification algorithm like BART is that of overfitting. It is possible that predictions are accurate because the data is rich and complex enough that it could predict any group of teachers. To rule this out, we run separate BART models for the following groups of countries: (1) a set of 4 countries chosen to have diverse characteristics (i.e., such that there should be limited to no capacity to accurately predict them; Australia, Belgium, Chile, Slovenia); (2) the countries from east Asia (Japan, Korea, Singapore, Shanghai); and (3) English-speaking countries (Australia, New Zealand, Alberta Canada, and the USA)[1]. In running these models, the same procedure was taken as for the focal results, but the countries included in the outcome variable were changed.

A comparison of models is shown in Figure A1, which presents the Receiver Operating Curve (ROC) for each model. ROC is a useful approach for comparing classification models (Han, 2022). They graph the false positive rate (FPR) by the true positive rate (TPR) for all cutoff values. A perfect model has an FPR of 0 and a TPR of 1, which corresponds to the top left corner. The closer a line is to the top left corner, the better the model. Figure A1 shows that the Asian model is the best model, followed by the Nordic model. Table A1 shows further details about the models, including the area under the curve (AUC) from the models, which measures the area under the curve from the ROC curves. AUC is considered a measure of overall model quality. Table A1 also shows classification rates by country in the target group for each model. The Asian model is the clear best model while the Nordic model is slightly, but not statistically significantly better than the English model. The diverse country model is by far the worst based on AUC values. Note that the confidence intervals for the AUC estimates were generated without accounting for the multilevel nesting of data, as interclass correlations were close to 0.05. Nonetheless, the confidence intervals may be biased to be slightly smaller than they should be. The online appendix shows the effective sample sizes for cases predicted to be near the cutoff, which were always over 1,000.

[1] Note that South Africa was also participating in TALIS, but they were dropped from English-speaking countries to test a reviewer's concern that the model captures wealthy western democracies. The English-speaking countries all now fall into this group.

While evidence for the Nordic model would have been most clearly provided if the Nordic model performed the best, the East Asian countries and the English-speaking countries (which are all former British colonies) are also recognizable groupings of countries with aspects of shared culture and educational systems. This sensitivity analyses, then, provides modest evidence for the Nordic model.

In principle, the ideal comparison case would be to generate a distribution of model fits for randomly selected countries. This, unfortunately, was not possible as the models take approximately a day to run, making it not practically possible to generate a distribution of random comparison models.

Figure A1: Receiver Operating Curve showing the results

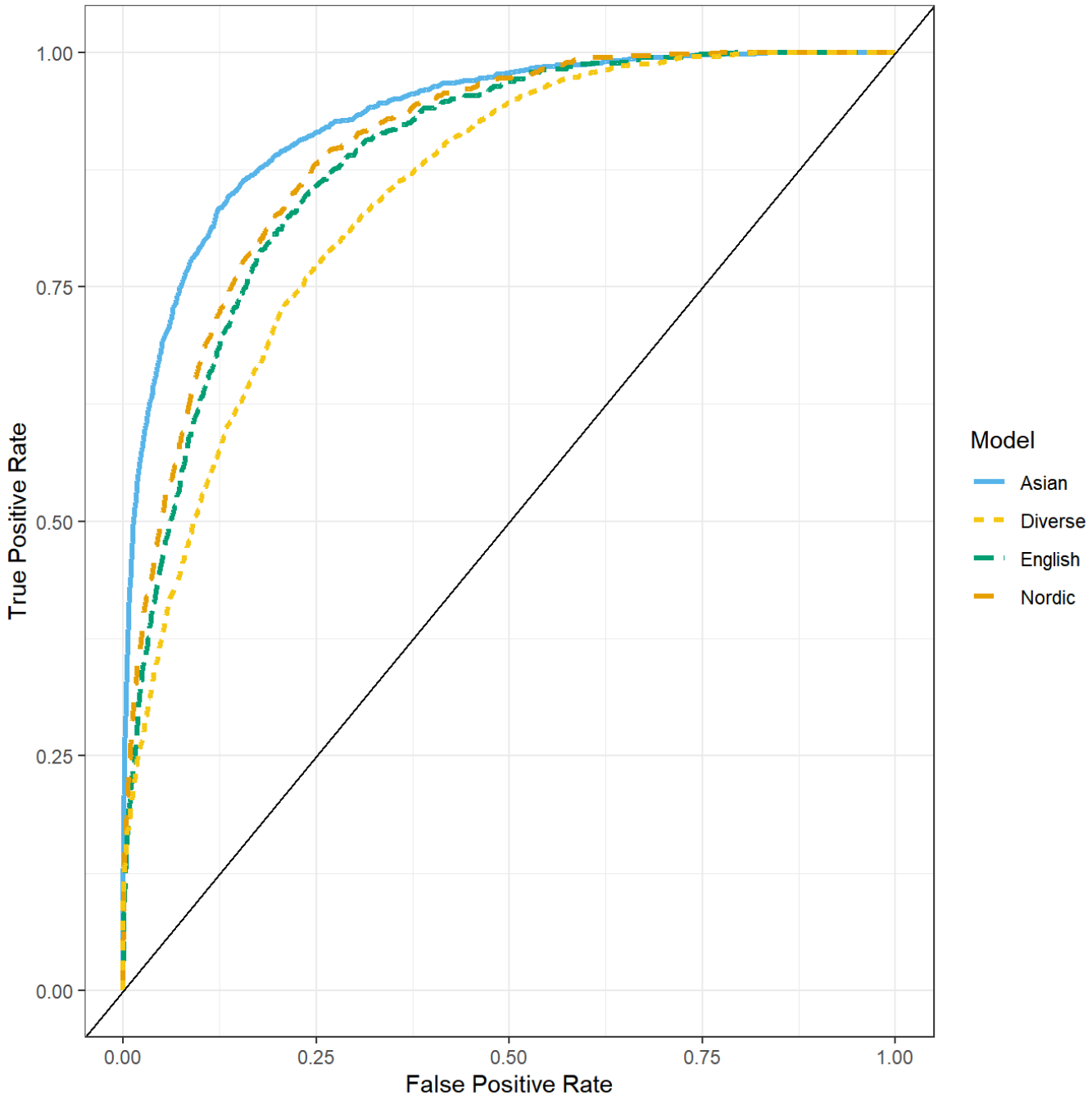


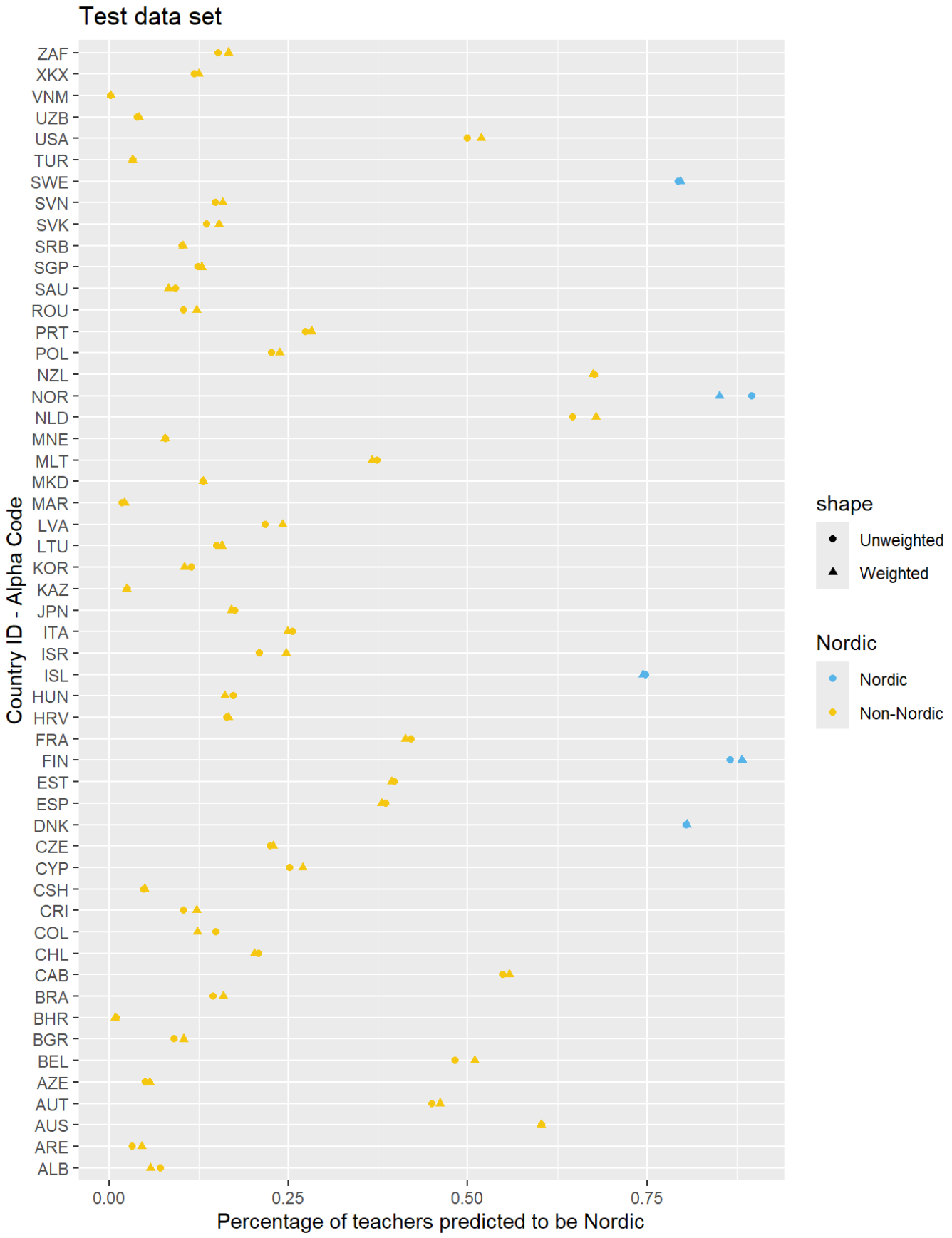
Table A1: Details on model quality

Model	AUC test data (85% CI)	Effective Sample Sizes	Youden Index (training data)	Youden cutoff (training data)	Sensitivity by country
Nordic Model	0.898 (0.888-0.907)	3044	0.633	0.074	DNK = 80% NOR = 90% SWE = 79% FIN = 87% ISL = 75%
English-speaking countries	0.886 (0.877-0.894)	3143	0.637	0.057	AUS =82% CAB =80% NZL =86% USA =76%
Diverse Countries	0.847 (0.840-0.853)	3370	0.542	0.075	AUS =72% CHL =71% SVN =91% BEL =75%
East-Asian Countries	0.930 (0.924-0.936)	3251	0.730	0.100	JPN =90% KOR =80% SGP =79% CSH =95%

Comparison of weighted and non-weighted models

The graph below shows the percentage of cases predicted to be Nordic for each country for using both weights on predicted probabilities and not weighting the predicted probabilities (which is how the data was used in the main paper). The figure shows that the predicted probabilities are generally the same, regardless of whether they are weighted or not. Where differences exist, they are quite small and not consistently in the same direction. In both cases, the Nordic teachers are clearly more likely to be predicted as Nordic than teachers from other countries. This would suggest that the results would remain equivalent whether or not they are weighted.

Figure A2: Predicted probabilities by country, weighted versus unweighted.


Test data set
ZAF
XKX
VNM
UZB
USA
TUR
SWE
SVN
SVK
SRB
SGP
SAU
ROU
PRT
POL
NZL
NOR
NLD
MNE
MLT
MKD
MAR
LVA
LTU
KOR
KAZ
JPN
ITA
ISR
ISL
HUN
HRV
FRA
FIN
EST
ESP
DNK
CZE
CYP
CSH
CRI
COL
CHL
CAB
BRA
BHR
BGR
BEL
AZE
AUT
AUS
ARE
ALB
Country ID - Alpha Code
0.00
0.25
0.50
0.75
Percentage of teachers predicted to be Nordic
shape
Unweighted
Weighted
Nordic
Nordic
Non-Nordic